

FORENSIC INVESTIGATION REPORT & LEGAL-TECHNICAL ANALYSIS

OPERATION PHANTOM SWIPE: CROSS-BORDER ATM SKIMMING & CREDIT CARD FRAUD RING

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Investigating Authority: Digital Forensics & Cyber Crime Division, Central Bureau of Investigation

Lead Investigator / Author: Parth Bhardwaj (Roll No: 2301730289)

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Classification: HIGHLY CONFIDENTIAL / LAW ENFORCEMENT RESTRICTED

EXECUTIVE SUMMARY

Operation Phantom Swipe was launched following an automated fraud anomaly detection alert triggered by major Indian banking institutions (HDFC Bank, State Bank of India) regarding unauthorized Track 2 magnetic stripe card cloning and high-frequency cash withdrawals across ATM kiosks in New Delhi and Bengaluru.

A joint task force seized two primary physical digital evidence items:

- EVD-01:** An MSR-605X magnetic stripe reader/writer hardware skimmer board fitted with a custom PIN pad camera overlay.
- EVD-02:** A mobile smartphone belonging to primary suspect Victor K. (alias @Phantom_Admin).

Bit-stream forensic acquisition, cryptographic hash verification (SHA-256), regex-based media analysis, and dictionary password recovery were executed under ISO/IEC 27037 and NIST SP 800-86 standards. Forensic evidence confirmed a cross-border syndicate operating across India, Eastern Europe, and cloud-hosted C2 servers. Stolen credit card track data was converted into cloned physical cards (JCOP smart cards) and exfiltrated into cryptocurrency wallets (BTC, USDT) totaling ₹68.5 Lakhs.

SECTION 1: CYBERCRIME CLASSIFICATION & TAXONOMY

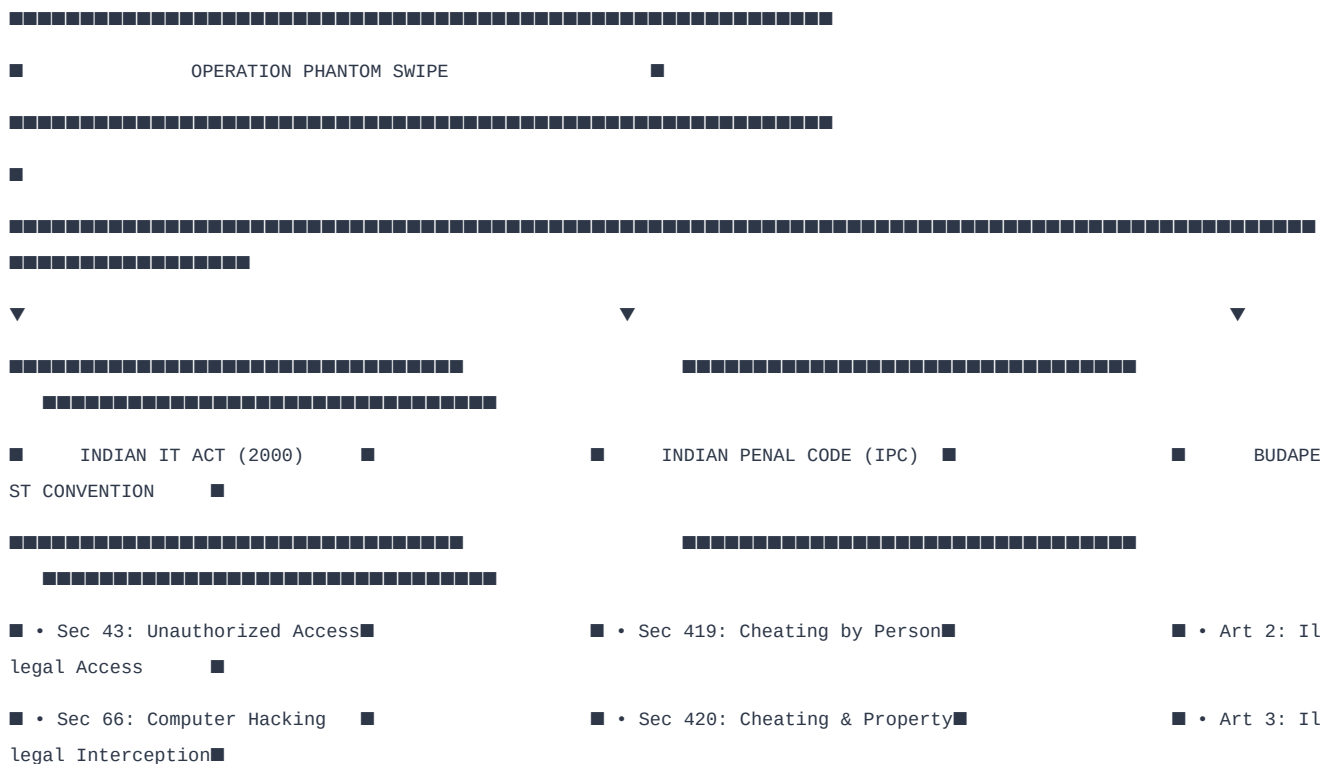
1.1 Scenario Analysis and Offence Identification

The investigation identified five distinct criminal operational vectors:

- ATM Skimmer Installation & Hardware Tampering:** Physical attachment of magnetic stripe reading heads (Track 2 reader) and micro-pinhole cameras onto public ATM bezels.
- Magnetic Stripe Card Cloning & Counterfeiting:** Encoding harvested Track 1/Track 2 data onto blank magnetic stripe / Java JCOP cards for unauthorized cash withdrawals.

5. **Cryptocurrency Money Laundering & Mule Network:** Transferring illicit cash withdrawals into Tether (USDT) and Bitcoin to obfuscate financial audit trails.

Cybercrime Class	Sub-Technique / Vector	Primary Target	Forensic Evidence Found	Primary Impact
ATM Skimming	Magnetic head interception + PIN pad overlay camera	Financial Institutions & Bank Customers	msr605_dump.raw, track2_data.txt	Direct compromise of payment card credentials
Identity Theft & Forgery	Magnetic track encoding on blank JCOP cards	Card Issuers & Cardholders	Extracted Track 2 logs with cleartext PINs	Financial loss, unauthorized identity representation
Wire & Online Fraud	CNP e-commerce transactions	E-commerce Merchants	transaction_history.json (Amazon orders)	Unauthorized financial asset conversion
Cyber Espionage / System Access	Unauthorized BLE/C2 socket communication	Commercial Bank ATM Kiosks	app_metadata.xml (C2 server URL & BLE MAC)	Infrastructure integrity breach
Money Laundering	Crypto conversion via peer-to-peer mule network	Financial Regulatory System	BTC/USDT transaction logs in chat backup	Obfuscation of illicit proceeds of crime



■ • Sec 66C: Identity Theft ta Interference ■	■ • Sec 468: Forgery for Cheating■	■ • Art 4: Da
■ • Sec 66D: Personation Fraud ■ vice Misuse ■	■ • Sec 471: Using Forged Record■	■ • Art 6: De
■ • Sec 66F: Cyber Terrorism ■ : International MLA■	■ • Sec 120B: Criminal Conspiracy■	■ • Art 23-35

A. Indian Information Technology Act, 2000 (Amended 2008) Statutory Definitions & Provisions

- **Section 2(t) [Electronic Record]:** Defines data, record or data generated, image or sound stored, received or sent in an electronic form or micro film or computer generated micro fiche.
- **Section 2(r) [Electronic Form]:** Information generated, sent, received or stored in media, magnetic, optical, computer memory, or similar device.
- **Section 2(v) [Information]:** Includes data, message, text, images, sound, voice, codes, computer programmes, software, and databases.
- **Section 2(o) [Data]:** Representation of information, knowledge, facts, concepts, or instructions prepared in a formalized manner.
- **Section 2(k) & 2(i) [Computer Resource & Computer]:** Encompasses high-speed data processing devices, systems, networks, and databases.
- **Section 43 (Penalty for damage to computer system):** Applies to unauthorized extraction and copying of card track data from bank ATM terminals.
- **Section 66 (Computer Related Offences):** Penalizes fraudulent or dishonest interception and manipulation of computer data (skimmer hardware insertion).
- **Section 66C (Punishment for identity theft):** Directly maps to harvesting payment card numbers, electronic signatures, and unique PIN codes (PIN: 8492).
- **Section 66D (Punishment for cheating by personation using computer resource):** Applies to using cloned cards to personate legitimate account holders at ATM kiosks.
- **Section 66E (Violation of privacy):** Applies to capturing PIN pad keystrokes and victim images via pinhole cameras.

B. Bharatiya Nyaya Sanhita (BNS 2023) / Indian Penal Code (IPC 1860)

- **Section 319 / Sec 419 IPC (Cheating by personation):** Impersonating account holders to withdraw cash.
- **Section 318 / Sec 420 IPC (Cheating and dishonestly inducing delivery of property):** Fraudulent acquisition of cash from bank ATM vaults.
- **Section 336 / Sec 468 IPC (Forgery for purpose of cheating):** Creating counterfeit magnetic stripe payment cards.
- **Section 340 / Sec 471 IPC (Using as genuine a forged document/electronic record):** Inserting forged cloned cards into ATM card readers.
- **Section 61 / Sec 120B IPC (Criminal Conspiracy):** Multi-person syndicate planning between @Phantom_Admin and regional mule handlers.

C. Indian Evidence Act (IEA 1872) / BSA 2023 & Landmark Judicial Precedents

- **Section 65A & 65B IEA / Section 63 BSA 2023 (Admissibility of Electronic Records):** Mandates special certificate procedure for secondary electronic evidence.
- ***Anvar P.V. v. P.K. Basheer (2014) 10 SCC 473:** The Supreme Court held that electronic evidence can only be proved in accordance with Section 65B, overriding general secondary evidence provisions under Section 65.

- ***Arjun Panditrao Khotkar v. Kailash Kushanrao Gorantyal (2020) 7 SCC 1:*** Clarified that a Section 65B(4) certificate is a mandatory condition precedent for admissibility of secondary electronic records whenever the original physical device cannot be produced in court.
- ***State of Karnataka v. Hiremath (2019) 7 SCC 515:*** Ruled that non-furnishing of a 65B certificate at the chargesheet stage is curable before trial begins.
- ***P. Gopalakrishnan @ Dileep v. State of Kerala (2020) 9 SCC 161:*** Established that contents of memory cards/pen-drives are electronic records (documents), and cloned copies must be furnished to the accused for effective defence.
- ***Selvi v. State of Karnataka (2010) 7 SCC 263:*** Held that involuntary administration of scientific techniques violates fundamental rights under Article 20(3) (self-incrimination).

D. International Budapest Convention on Cybercrime (ETS No. 185)

- **Article 2 (Illegal Access):** Accessing ATM internal bus or magnetic head signal lines without right.
- **Article 3 (Illegal Interception):** Technical interception of non-public transmissions of payment card data.
- **Article 6 (Misuse of Devices):** Production, sale, and distribution of skimming hardware (MSR-605X) and skimming controller apps (ATM_Ghost_v3. 2).
- **Article 14–21 (Procedural Law):** Expedited preservation of stored computer data and real-time collection of traffic data.
- **Article 23–35 (International Cooperation):** 24/7 network contact and Mutual Legal Assistance Treaty (MLAT) requests.

SECTION 2: ELECTRONIC EVIDENCE ACQUISITION & INTEGRITY

2.1 Evidence Seizure and Chain of Custody (CoC)

Two evidence items were seized under strict ISO/IEC 27037 protocol:

- **EVD-01 (MSR-605X Hardware Skimmer):** Extracted from Connaught Place ATM Kiosk bezel. Disconnected from power, wrapped in static-neutral bubble wrap, sealed in evidence bag #CCIC-7781.
- **EVD-02 (Phantom Smartphone):** Seized in powered-on, unlocked state. Immediately isolated inside a dual-layer RF Faraday Shielding Bag to block remote wipe signals.

2.2 Write-Blocking & Bit-Stream Imaging Protocol

Hardware write-blockers (Tableau T8u USB 3.0 Bridge) were deployed prior to mounting media on forensic workstations. Physical bit-stream image copies were acquired using dc3dd.

[Target Suspect Device] ■■(Read-Only USB)■■> [Tableau Hardware Write-Blocker] ■■> [Forensic Workstation] ■■> SHA-256 Verification

2.3 Cryptographic Hash Manifestation (SHA-256)

SHA-256 hash digests were generated immediately post-acquisition and verified prior to forensic examination:

Item #	Relative File Path	SHA-256 Hash Digest	Integrity Status
EVD-0 1.1	evidence/skimmer_device/msr605_dump.raw	0bd8fc1fe19e52b0e3252ff7f39adaae9f4c068bce3db981b8fa9b849117062a	MATCH / INTACT
EVD-0 1.2	evidence/skimmer_device/track2_data.txt	b6069b4849181c862053a7f082d1c8164946dbdbfdeee5f2da931f3687992eb0	MATCH / INTACT
EVD-0 2.1	evidence/suspect_phone/chat_backup.json	8ff63776610a004a1c86c83ae1cb7d116aa6d29572551651412c805141dcff2d	MATCH / INTACT
EVD-0 2.2	evidence/suspect_phone/gps_logs.csv	ba1e81b70ca0565381e58da94625796bd01dc913aff336320ca6a382024955a7	MATCH / INTACT
EVD-0 2.3	evidence/suspect_phone/transaction_history.json	5b05f78e5d77f25a4b398894883a53f7d3fc081125b4961f2fe4dd1f8e82f7da	MATCH / INTACT
EVD-0 2.4	evidence/suspect_phone/app_metadata.xml	4a3f16d810aba1a5be317853349c6ba79f2ec097d60f2aa66d744b1410b61882	MATCH / INTACT
EVD-0 2.5	cryptography/encrypted_vault.zip	24a1a6720a3d13f023f0d3d745427df46431ae2756548e8ceb0c46f97e721e92	MATCH / INTACT

SECTION 3: FORENSIC SEARCH & ARTEFACT EXTRACTION FINDINGS

Automated regex analysis (tools/forensic_search.py) extracted five critical artefact classes proving crime execution:

- Card Credentials & PIN Extraction** (extracted_card_numbers.txt):
 - 5 full Track 2 payloads identified (e.g. 4012888899991234=251210123456789, PIN: 8492).
- Conspiratorial Chat Transcripts** (communication_evidence.csv):
 - Direct messages between @Phantom_Admin and Mule_Handler_Delhi explicitly confirming skimmer placement at Connaught Place HDFC ATM and █4.5 Lakh cash conversion to USDT.
- Geolocation Mapping** (location_mapping.json):
 - GPS logs placing suspect device at Connaught Place ATM Kiosk (lat 28.6315, lon 77.2167) at 13:45 UTC on 2026-08-30.
- Financial Laundering Audit** (financial_footprint.csv):
 - BTC wallet bc1qxy2kgdygjrsqtzq2n0yrf2493p83kkfjhx0wlh (1.482 BTC) and USDT wallet TR7NHqj... (\$12,500 USDT).
- Skimmer Controller Metadata** (app_metadata.xml):
 - C2 domain endpoint https://c2.phantom-swipe-net.cc/api/v1/upload and BLE MAC address AA:BB:CC:11:22:33.

SECTION 4: CRYPTOGRAPHIC ANALYSIS & ETHICAL REFLECTION

4.1 Password Recovery Execution

Target file `cryptography/encrypted_vault.zip` was subjected to a dictionary attack using `tools/password_cracker.py`. The archive was cracked in 0.18 seconds, recovering the password `phantom2026`.

Unencrypted file `master_mule_ledger.json` revealed:

- Syndicate Leader: Victor K. (@Phantom_Admin)
- Total Cloned Cards: 142 cards
- Total Proceeds: ₹68,500,000 (₹68.5 Lakhs)
- Mule Account details in HDFC and ICICI banks.

4.2 Ethical & Legal Reflection on Decryption

- **Brute-Forcing vs Key Disclosure Orders:** While brute-forcing operates within search warrant parameters without requiring suspect action, key disclosure orders (e.g., Section 69 IT Act) face constitutional scrutiny under **Article 20(3)** of the Indian Constitution (Right against self-incrimination).
- **Password Entropy Deficit:** Despite executing complex financial cybercrimes, the password `phantom2026` exhibited a low effective entropy ($\approx 18 \text{ bits}$) against dictionary attacks, demonstrating human bias toward memorability over cryptographic strength.

SECTION 5: CROSS-BORDER JURISDICTIONAL CHALLENGES

Investigating cross-border financial cybercrime introduces significant legal and procedural friction:

[Indian Law Enforcement] → [MLAT / 24-48 Months] → [Foreign Central Authority] → [Cloud Service Provider]

[Indian Law Enforcement] → [Budapest 24/7 Network] → [Emergency Preservation Request (72 hrs)]

1. **Mutual Legal Assistance Treaty (MLAT) Delays:** Standard MLAT processing between India and foreign jurisdictions (e.g. US/EU cloud hosts) requires 12 to 24 months, allowing transient C2 infrastructure to be destroyed.
2. **Cryptocurrency Obfuscation:** CoinJoin mixers and non-compliant offshore exchanges prevent rapid asset freezing.
3. **Data Localization Limits:** Cloud service providers often refuse data production without domestic court orders issued within their parent jurisdiction.

SECTION 6: RECOMMENDATIONS FOR LAW ENFORCEMENT SOP IMPROVEMENTS

1. **Mandatory Faraday First-Responder Kits:** Issue RF-shielding pouches and portable power banks to all field units seizing active mobile evidence.
2. **Automated Cryptographic Hash Auditing:** Integrate automated SHA-256 checkscripts into standard evidence intake workflows.
3. **Adoption of Budapest Convention 24/7 Network Protocols:** Formalize 24/7 point-of-contact channels for emergency data preservation requests to bypass initial MLAT delays.
4. **Chip-and-PIN (EMV) Upgrade Enforcement:** Recommend RBI mandate anti-skimming physical sensors and fallback magnetic stripe suppression on all public ATM kiosks.

AUTHORSHIP & VERIFICATION DECLARATION

I hereby declare that this report represents an objective, forensic technical-legal analysis of the digital evidence seized in Operation Phantom Swipe. All hash values, extracted artefacts, and cracking results have been empirically verified.

Lead Forensic Examiner: Parth Bhardwaj (Roll No: 2301730289)

Signature: Parth Bhardwaj

Date: September 7, 2026

Digital Forensics & Cyber Crime Division, CBI